

Branch `refit-and-semantics-guard`. Suite **56 passed, 1 failed, 0 skipped** (from 54/1 at the start), and the suite is now genuinely strict for the first time.

Every one changed under measurement. That is the headline, and it is a pattern rather than three coincidences.

item	what was claimed	what was measured
1. redirection is most of a 23% drop	"probably most of it"	1.60%
2. the 18 pair weights are collinear	nine flipped sign, so unstable	max VIF 2.2 , zero flips on 23% more data
3. self-play teams are not the metagame	TVD 54.8%, every top species under	3.0-point mean gap, four of twelve OVER

Item 1's premise is void. The protocol emits no `-activate` line for a redirect and prints only a move's RESOLVED target, so redirection cannot make a click unmatchable — the redirector is a legal candidate, the matcher finds it and accepts the click *with the wrong label*. That is real and small: 1.55% of clicks, and unrecoverable from the store because the chosen target was never written down.

What the 23% actually was: the foe switched in on the same turn (44.4% — a human aims at a SLOT, the store records a SPECIES, and switches resolve first), an in-battle forme change with no sheet entry (19.7%), a mirror collapsing the two team sheets (16.4%), and no target recorded at all (13.7%). All fixed in `engine/click_match.js`: usable joint turns **76.80% → 94.52%**.

Item 3's 54.8% could not be reproduced by any tool in the repository, and the metric behind it is not recorded anywhere. The deduplicated team pool is a *documented deliberate choice* (`mew.js:270`), not an oversight. The large realism gaps are mechanical, in the category that report labels "fix these": Protect is 23.2% of MAG's moves against a human 13.8%.

- **Both vectors refitted** on the corrected matcher and the corrected damage table. 176,981 → **199,472** usable decisions; joint turns 66,236 → **82,065**.
 - **The marginal vector won its head-to-head:** 58.4% of 238 decisive pairs, DECIDED at 219 by `sprt.js`, `paired_h2h.js` agreeing 139/139 and 99/99. Stopped at 2,750 games of 20,000.
 - **Greedy still pays** on today's vector: 79.7% of decisive pairs, decided after 55.
 - `build/omnibus.py` **produces PDFs again** — Chrome → Edge → LibreOffice, discovered not hardcoded, zero-byte output treated as failure.
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Will: "each time I tell you to research it and implement a universal fix, and each time you devise more and more tests and they continually fail. Are we actually making progress or going in circles?"

The number said circles: error-swallowing blocks went **233 → 238** across the session that was meant to be fixing them. Every guard MEASURED the disease; none cured it.

The disease is the return contract. `lookup(x) -> null` means both "genuinely not in the data" and "you asked the wrong question", and those are indistinguishable at the call site. A guard per pathway cannot finish because there is no end to the pathways.

`engine/lookup.js`: **a miss throws unless the caller declares it, with a reason.** The reason is required rather than a bare `true` — a boolean is not a parameter, and the reason is what a reviewer needs. It also makes the opt-outs greppable, so the number of places tolerating a miss is a quantity that can be driven DOWN, unlike a test count.

There are six declared misses in the entire codebase. That is the number to watch.

Eight species had no damage-table row, so every damage-derived feature read ZERO for them:

- **five format megas** — Victreebel, Feraligatr, Skarmory, Barbaracle, Falinks-Mega. The data existed all along in `mega-dex-official.json`, gated behind `in_our_store`, a flag measured on 2026-07-24 against a smaller store. A stale boolean silently withholding real data.
- **three formes** — Aegislash-Blade, Palafin-Hero, Gourageist's sizes. These need real rows: their stats differ from the base, so the cosmetic fallback *correctly* refuses to substitute.

Aegislash-Blade is the sharp one: 140 Attack against Shield's 50, so MAG was blind to it in exactly the state where it is dangerous.

`board.dmgMon.unknownSpecies` 6.16% -> 3.62% -> 0.00%

The degradation budget was certifying a workload, not the world. It read 0.00% with a 0.00% ceiling — correct for 120 REPLAYED games, in which an in-battle forme change never reaches a scored position. Adding 60 self-play games measured 6.16%. R7 for the third time in two days.

The semantics guard has been a warning since the day it was written. `ABRA_STRICT_SEMANTICS=1` makes stale weights fatal; `magnemite.js` implements it and its own comment says "*which is what tests/run-all.js and any fit should use*" — **and no file in this repository ever set it.** Proved rather than assumed: I made the fixture genuinely change, it printed 22 features whose meaning had moved, and then loaded the weights anyway. `tests/run-all.js` now sets it. Live play keeps the warning, and that trade-off stays argued where it is made.

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- **"Sucker Punch's 48% failure prices the missing opponent model."** It does not. With sampling instead of greedy: MAG 33.2%, human 33.9%, at identical usage. Greedy does all of it — it takes the argmax every time, so a condition-dependent move gets clicked in every spot where it whiffs. I said this before running the test that settled it, which is the exact failure this session was about.
 - **My first fixture fix was INERT.** Benched Aegislash where `benchRisk` is identically zero, so it moved no number — coverage that looks improved and is not.
 - **My second REGRESSED coverage.** Made it active, displacing the Fire partner Torkoal's sun boosts; `weatherSetupHelpsPartner` then fired nowhere.
 - **Even the third was subtly wrong.** Benching *Aegislash* exercises the Shield row, which was never missing — Aegislash-Blade only exists once Stance Change flips it, which a static fixture never does. Deleting the `aegislash-blade` row moved ZERO hashes. It took **Gourageist-Super**, a species you actually bring, for the guard to bite.
 - **I restamped policy-weights-nopop.json when it needed a refit.** A restamp is only valid when the feature FUNCTION is unchanged; that file was fitted against the original 308-species table. I papered over real staleness a few hours after fixing the identical defect in `build_mag_variables.js`.
 - **My own isNonstandard bundle filter changed behaviour.** `board.js` SWEEPS the move table to derive sets, so filtering cut `STALL` from 11 moves to 6 and `tgtMayProtect` scored every Pokemon holding Silk Trap as one that never blocks. Caught by the test I had just written.

Every one of these was settled by RUNNING something, not by reasoning about it.

`tests/test-mag-page` — `app/index.html` carries a SECOND feature scorer assigning 21 of 56 features, so a site reader sees numbers the bot did not compute.

The blocker everyone assumed was fundamental is gone. `board.js` was believed unable to run in a browser because it CALLS dex handlers (`onModifyType`, `onModifyPriority`, `basePowerCallback`) and functions do not survive JSON. Measured: `board.js` invokes every one with a `this` context IT builds, so they

close over nothing — **225 of 225 rehydrate through** `toString()` . `board.js` now runs in a browser and `tests/test-board-browser.js` proves it scores **56 of 56 features identically to node**.

What remains is not engineering, it is a cost decision. Swapping the page's scorer needs:

file	KB	already loaded?
<code>data/board-data.js</code>	817	no
<code>data/abra-tags.js</code>	383	no
<code>engine/board.js</code>	193	no
<code>data/move-effects.js</code>	128	no
<code>engine/medicham2-browser.js</code>	85	no
<code>engine/mc_key.js</code> + <code>tags.js</code>	15	no

~**1.4 MB added to a page currently ~460 KB**. Three options, in my order of preference:

1. **Lazy-load the scorer when the MAG room is opened.** The page stays 460 KB; only a visitor who actually opens that room pays. Correctness with no cost to anyone else. This is the right answer and it is the one I would do.
2. Ship it eagerly and accept 4x page weight.
3. Leave the page as it is and mark the test as a known gap — the status quo, in which the site shows numbers the bot did not compute.

The load order is load-bearing and was measured: raw `abra-tags.js` → `medicham2-browser.js` (which snapshots the raw table) → `engine/tags.js` (which replaces the global with the accessor) → `board.js` . Getting it wrong degrades silently in both directions and cost ten of the twelve disagreements before it was found.

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1. **The page swap**, once you have picked an option above. Everything else is done.
 2. **~~Greedy over-clicks conditional moves.~~ WITHDRAWN 2026-08-02 evening.** The 47.9/33.2/33.9 triple came from two corpora differing on THREE levers (greedy, joint, weights file). On a single-lever corpus the gap is -0.7 points against a measured known-zero band of [-7.5, +7.0]. See `docs/SESSION-2026-08-02-evening.md` §1.
 3. **~~Protect's other half.~~ RESTATED.** The rate is real (sampling 1.24x human, not 1.44x) but it is not in the weights: the shipped vector predicts 19.22% against an observed 18.86% on the corpus it was fitted on. The live finding is the SPECIES-CONDITIONAL gap underneath that average, +12 to -19 points. See §2.
 4. **~~MAG under-switches.~~ WITHDRAWN as stated.** The `partial_bring` evidence was circular, and the filter is a decisiveness filter — the short side is the WINNER in 1,008 of 1,008 games. The real defect was that `switching: false` in every corpus on disk, so voluntary switches were exactly zero. Turned on it wins 54.1% of 1,032 decisive pairs. See §3.
 5. **Extend the lookup contract to the remaining accessors.** `mcKey` and `loadData` are done. The sheet index and the tag tables are the obvious next two.
 6. `terrainSetupHelpsPartner` **carries the pair block's largest weight on a 0.00% fire rate.** Three pair weights are barely observed; `data/collinearity-joint.json` records the fire rate beside every one so they cannot be confused with well-supported ones.
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Seven bugs this session, all one shape: **nothing crashed**. Every one returned a plausible value and carried on. The forme lookup, the mirror sheet, the switched-in target, the browser tags, the move filter, the alias map, the stale mega flag — the counters were right and nothing failed.

And separately, six *claims* had the same shape: redirection as the cause, the 54.8%, the no-popularity column, `opponent_recall` 's hardcoded staleness string, `build_board_browser` 's claim that a test existed, and my own Sucker Punch diagnosis. A true-sounding sentence written down as a literal cannot go wrong loudly — only quietly, and only later.

The countermeasure is the same for both: **derive it, or measure it — and if you must write it down, write the thing that checks it.**